# Large Scale Determination of Permeability-Relevant Lipophilicity and Reaction Profiling in DNA-Encoded Libraries - Algorithms

## Supplementary Information

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| | |
|---|---|
| CC | Classic Chrome |
| GPPM | Gaussian Peak Picking Model |
| HGPPM | Hierarchical Gaussian Peak Picking Model |
| STA | Split-Tree Analysis |
| SWM | Sliding Window Minimum |

# 4 Introduction

This supplement provides a comprehensive formalized description of the algorithms developed for the identification of product peaks from sequence count chromatographic time series data and library reaction profiling visualization. For more details on how the code works, please see the GitHub repository. The algorithms discussed herein are:

1. Gaussian Peak Picking Model (GPPM)
    - Hierarchical Gaussian Peak Picking Model (HGPPM)
2. Classic Chrome (CC)
3. Split-Tree Analysis (STA)

These algorithms are designed to analyze chromatographic data, which represents the temporal separation of chemical compounds. The GPPM represents our initial approach, while CC represents a more advanced iteration in algorithm design, incorporating additional data from null compounds to construct a null truncation hierarchy.

The codebase has been written in a modular fashion such that the modules of each algorithm can be interchanged. HGPPM represents the combination of CC and GPPM to demonstrate this modularity. Furthermore, the codebase has been designed to facilitate easily customized configurations of each module and the extension of base classes to implement user-defined versions. Finally, STA can be used with the outputs of any configuration or with otherwise pre-picked peak data.

# 5 Chromatogram Peak Picking

Peak picking by the canonical algorithms (CC, GPPM, HGPPM) is achieved through several analysis, processing, and preprocessing steps:

1. Input
2. Chromatogram Analysis
3. Baseline Correction
4. Initial Peak Finding
5. Peak Analysis
6. Processing
7. Peak Selection
8. Output

The input, chromatogram analysis, initial peak finding, peak analysis, and output steps are shared between all algorithms while they differ in their baseline correction methods and processing step. The canonical information flow through these steps are shown in Fig. 1. Also included is the abstract customizable flows which are possible via user-defined configurations.

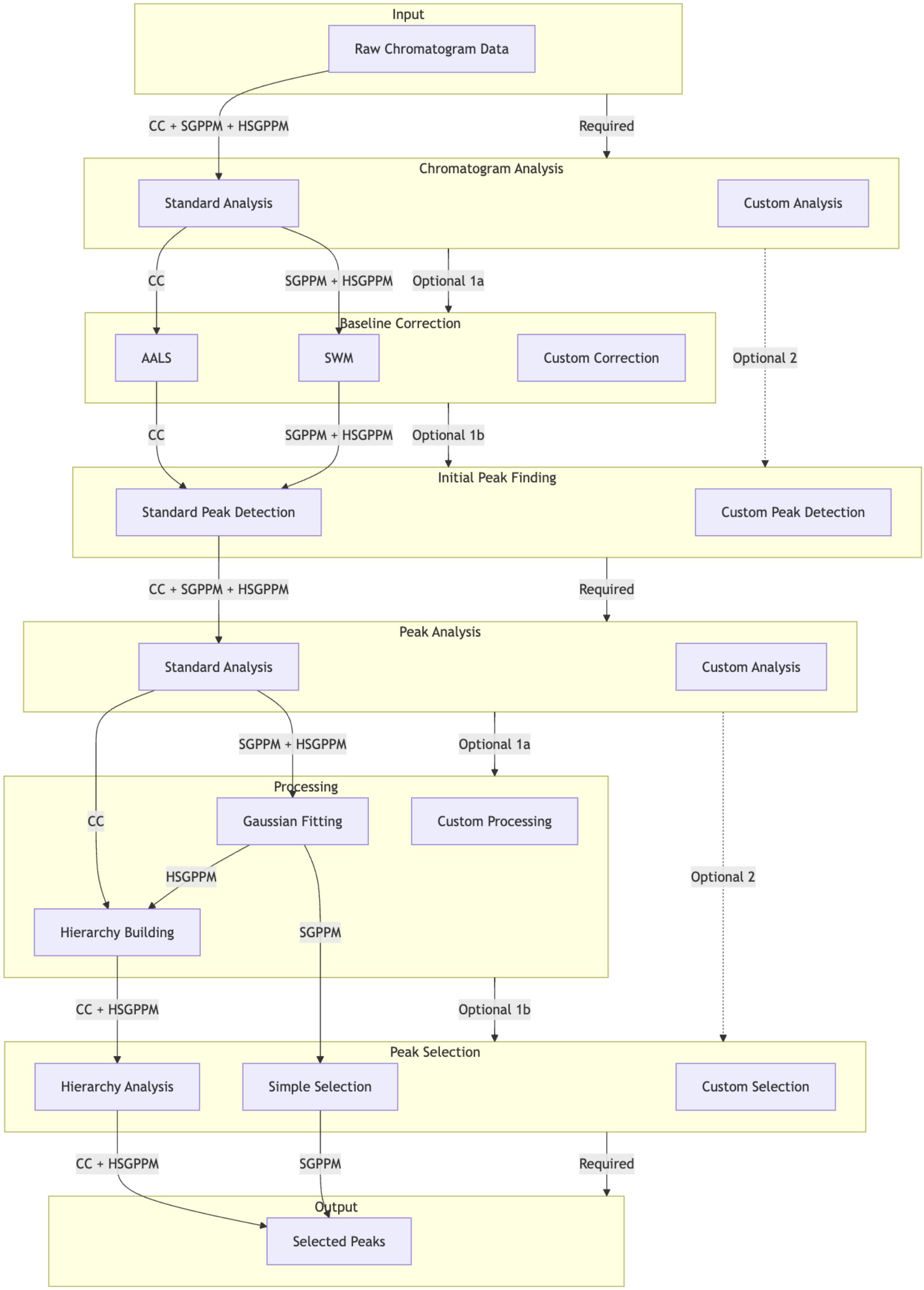


Fig. 1. Information flow through CC, GPPM, and HGPPM algorithms as well as modular custom paths.

## 5.1 Chromatogram Analysis

Several metrics are computed on the raw chromatogram signal to characterize signal quality, noise characteristics, baseline properties, and overall distribution. Define the following variables as such:

- Let $c$ be a chromatogram such that $c \in C$ where $C$ is the set of all chromatograms
- Let $i$ be the index of the $i^{th}$ chromatogram in $C$
- Let $t$ be the time index of a given signal
- Let $(y|i,t)$

Let $C$ be the set of chromatograms $c \in C$. Let $c_i^x(t)$ represent the time values and $c_i^y(t)$ and the height values of the $i^{th}$ chromatogram in $C$ at time $t$ where $t \in [1, c_i^n]$. These metrics are as follows:

- Noise Metrics
  - Moving Mean $\breve{\mu}\,(y|i,t,w,\delta)$:

$$\breve{\mu}\,(y|i,t,w,\delta) := \frac{1}{w} \sum_{j \in [-\delta,\delta]} c_i^y(t+j) \tag{1}$$

  - Moving Standard Deviation $\breve{\sigma}\,(y|i,t,w,\delta)$:

$$\breve{\sigma}\,(y|i,t,w,\delta) \tag{2}$$

$$c_i^{\breve{y}}(t) = \sqrt{\frac{1}{w-1} \sum_{j \in [-\delta,\delta]} \left[c_i^y(t+j) - c_i^{\tilde{y}}\right]^2} \tag{3}$$

    Computed at each point for window length $w$ and $\delta = \left\lfloor \frac{w}{2} \right\rfloor$ where $c_i^{\tilde{y}}$ is the is the local mean in the window around $t$.

  - Minimal Variation Regions ($c_i^M$): Each region $[t_s, t_e]$ satisfies:

$$\begin{aligned} c_i^{\breve{y}}(t) &< \theta \cdot \min_{t \in [1, c_i^n]} c_i^{\breve{y}}(t) \; \forall \; t \in [t_s, t_e] \\ t_e - t_s &\geq w_{min} \\ t_s &\geq c_i^n \cdot \epsilon \\ t_e &\leq c_i^n(1-\epsilon) \end{aligned} \tag{4}$$

    where:
    * $t_s$ is the start time of the region
    * $t_e$ is the end time of the region
    * $\theta$ is the variation threshold
    * $w_{min}$ is the minimum region width
    * $\epsilon$ is the edge exclusion fraction

  - Noise Level ($c_{i,\sigma_n}$): median, denoted $P_{50}$, of standard deviations in quiet regions:

$$c_{i,\sigma_n} = P_{50}\{c_{i,\sigma_m}(t) : t \in c_{i,M}\} \tag{5}$$

  - Signal to Noise Ratio (SNR):

$$c_{i,\text{SNR}} = \frac{\max_{t \in [1,n]} c_{i,y}(t) - \min_{t \in [1,n]} c_{i,y}(t)}{c_{i,\sigma_n}} \tag{6}$$

- Baseline Metrics

– Baseline Mean ($c_{i,\mu_b}$): Estimated using the 10th percentile , denoted $P_{10}$, of signal values:

$$c_{i,\mu_b} = P_{10}\{c_{i,y}(t)\} \ \forall \ t \in [1, n] \tag{7}$$

– Baseline Drift ($c_{i,\beta}$): First-order coefficient from linear regression:

$$c_{i,\beta} = \frac{n \sum_{t=1}^{n} [t \cdot c_{i,x}(t) \cdot c_{i,y}(t)] - \sum_{t=1}^{n} [t \cdot c_{i,x}(t)] \cdot \sum_{t=1}^{n} [c_{i,y}(t)]}{n \sum_{t=1}^{n} [t^2 \cdot c_{i,x}(t)^2] - \sum_{t=1}^{n} [t \cdot c_{i,x}(t))^2]} \tag{8}$$

- Area Metrics

  – Total Area ($c_{i,A}$): Computed using trapezoidal integration:

$$c_{i,A} = \frac{1}{2} \sum_{t=1}^{n-1} [c_{i,x}(t+1) - c_{i,x}(t)] [c_{i,y}(t+1) + c_{i,y}(t)] \tag{9}$$

  – Positive and Negative Areas ($c_{i,A^+}$, $c_{i,A^-}$): Separated by sign:

$$\begin{aligned} c_{i,A^+} &= \frac{1}{2} \sum_{t=1}^{n-1} [c_{i,x}(t+1) - c_{i,x}(t)] [\max\{c_{i,y}(t+1), 0\} + \max\{c_{i,y}(t), 0\}] \\ c_{i,A^-} &= \frac{1}{2} \sum_{t=1}^{n-1} [c_{i,x}(t+1) - c_{i,x}(t)] [\min\{c_{i,y}(t+1), 0\} + \min\{c_{i,y}(t), 0\}] \end{aligned} \tag{10}$$

  noting that $c_{i,A^-}$ will typically always be 0 for raw chromatographic data. It is implemented purely for completeness, possible future applications, and as a sanity check.

- Distribution Metrics

  – Skewness ($c_i^\gamma$): Third standardized moment:

$$c_i^\gamma = \frac{1}{c_i^n \sigma^3} \sum_{t=1}^{c_i^n} \left[c_i^y(t) - c_i^{y_\mu}\right]^3 \tag{11}$$

  – Kurtosis $\left(c_i^{(\kappa)}\right)$: Fourth standardized moment:

$$\kappa = \frac{1}{c_i^n \sigma^4} \sum_{t=1}^{c_i^n} [y(t) - \bar{y}]^4 - 3 \tag{12}$$

  – Dynamic Range ($R$):

$$R = \max_{t \in [1,n]} y(t) - \min_{t \in [1,n]} y(t) \tag{13}$$

- Quality Metrics

  – Signal Smoothness ($S$): Mean absolute first derivative:

$$S = \frac{1}{n-1} \sum_{t=1}^{n-1} \left|y(t+1) - y(t)\right| \tag{14}$$

– Baseline Roughness ($\rho$): Standard deviation of first derivative:

$$\rho = \sqrt{\frac{1}{n-2}\sum_{t=1}^{n-1}\left[y(t+1) - y(t) - \bar{d}\right]^2} \tag{15}$$

where

$$\bar{d} = \frac{1}{n-1}\sum_{t=1}^{n-1}\left[y(t+1) - y(t)\right] \tag{16}$$

is the mean first derivative

## 5.2 Baseline Correction

There are two canonical baseline correction algorithms which are used by the algorithms:

- Sliding Window Minimum (SWM)
- Adaptive Asymmetric Least Squares (AALS) [?]

### 5.2.1 Sliding Window Minimum

Given an original signal $y(t)$, we first extend the original signal $y(t)$ defined on $[1, n]$ to a padded signal $y_{pad}(t)$ on $[-\delta, n+\delta]$ where the window length is $w$ and $\delta = \lfloor \frac{w}{2} \rfloor$:

$$y_{pad}(t) = \begin{cases} y(1), & t \leq 0 \\ y(t), & 1 \leq t \leq n \\ y(n), & t > n \end{cases} \tag{17}$$

Next, compute the SWM baseline $z(t)$:

$$z(t) = \min_{s \in [-\delta, \delta]} \{y_{pad}(t+s)\} \tag{18}$$

Then obtain the corrected signal $y_{corr}(t)$:

$$y_{corr}(t) = \max\{y(t) - z(t), 0\} \tag{19}$$

### 5.2.2 Adaptive Asymmetric Least Squares Baseline Correction

Given an original signal $y(t)$, the baseline $z(t)$ is estimated by minimizing [?]:

$$S = \sum_{t=1}^{n} w(t)\left[y(t) - z(t)\right]^2 + \lambda \sum_{t=1}^{n}\left[\Delta^2 z(t)\right]^2 \tag{20}$$

where:

- $S$ is the objective function to be minimized
- $w(t)$ are asymmetric weights
- $\lambda$ is a smoothing parameter (typically $10^2 \leq \lambda \leq 10^9$)
- $\Delta^2 z(t) = z(t) - 2z(t-1) + z(t-2)$ is the second-order difference of $z(t)$

The weights are updated iteratively according to:

$$w(t) = \begin{cases} p & \text{if } y(t) > z(t) \\ 1-p & \text{if } y(t) \leq z(t) \end{cases} \tag{21}$$

where $p$ is the asymmetry parameter ($0.001 \leq p \leq 0.1$). The system of equations to be solved at each iteration is:

$$(W + \lambda D'D)z = Wy \tag{22}$$

where $W = \text{diag}(w)$ and $D$ is the difference matrix such that $Dz = \Delta^2 z$. As stated in the original report of AALS, supported by our observations, the algorithm seems to typically converge within 5-10 iterations. The baseline-corrected signal $y_{\text{corr}}(t)$ is then obtained by:

$$y_{\text{corr}}(t) = y(t) - z(t) \tag{23}$$

The smoothing parameter $\lambda$ controls the flexibility of the baseline, while $p$ determines the asymmetry of the fitting. Parameters should be selected by testing the algorithm on a few sample chromatograms, though using the default configuration may be sufficient.

### 5.3 Initial Peak Finding

Initial peak finding is done using SciPy's `find_peaks` method [?]. Before `find_peaks` is called, we compute a height threshold $T_h$, window length threshold $T_l$ , prominence threshold $T_p$, and width threshold $T_w$ which are passed into `find_peaks`. A relative height threshold $T_r$ is also passed in via the `PeakFinderConfig`.

The initial set of peaks $P_i$ are identified as local maxima in $y_{corr}(t)$:

$$P_i = \{p_t : y_{corr}(p_t - 1) < y_{corr}(p_t) > y_{corr}(p_t + 1)\} \tag{24}$$

where $p_t$ is a peak at time $t$. The subset of peaks $P_h \subset P_i$ found by filtering $P_i$ by height threshold $T_h$:

$$P_h = \{p_t : y_{corr}(p_t) > T_h, p_t \in P_i\} \tag{25}$$

The subset of peaks $P_p$ is found by considering a horizontal line at height $y_{corr}(p_t)$ extending in both directions until either the signal is intersected (ignoring points which are at the same height as the horizontal line) or until the line is of length $T_l$. Let $p_{t_p}$ be the prominence of peak at time $t$. The set $P_p \subset P_h$ is given by:

$$P_p = \{p_t : p_{t_p} > T_p, p_t \in P_h\} \tag{26}$$

noting that, during the prominence calculation, the left and right peak bases are found, $p_{t_{BL}}$ and $p_{t_{BR}}$ respectively, given by:

$$\begin{aligned} p_{t_{BL}} &= \underset{t \in [p_t - T_l, p_t]}{\arg\min} \{y_{corr}(t) : y_{corr}(t') \leq y_{corr}(p_t) \ \forall \ t' \in [t, p_t]\} \\ p_{t_{BR}} &= \underset{t \in [p_t, p_t + T_l]}{\arg\min} \{y_{corr}(t) : y_{corr}(t') \leq y_{corr}(p_t) \ \forall \ t' \in [p_t, t]\} \end{aligned} \tag{27}$$

The subset of peaks $P_w$ is found by drawing a horizontal line in the same fashion as with finding the peak prominence, though, instead of using $y(p_t)$ as the horizontal line height, the height $h_{eval}$ of the line is computed as:

$$h_{eval} = y_{corr}(p_t) - T_r \cdot p_{t_p} \tag{28}$$

In order to achieve a more accurate width calculation, the corrected signal is first interpolated. Given the interpolated signal $y_{interp}(t)$, prominence $p_{t_p}$, and peak bases $p_{t_{BL}}$ and $p_{t_{BR}}$ for each peak, we first find the intersection points:

$$\begin{aligned} p_{t_{I_L}} &= \max\{t : y_{interp}(t) = h_{eval}, p_{t_{BL}} \leq t < p_t\} \\ p_{t_{I_R}} &= \min\{t : y_{interp}(t) = h_{eval}, p_t < t \leq p_{t_{BR}}\} \end{aligned} \tag{29}$$

the peak with $p_{t_W}$ is given by:

$$p_{t_W} = p_{t_{I_R}} - p_{t_{I_L}} \tag{30}$$

Then, $P_w \subset P_p$ is given by:

$$P_w = \{p_t : p_{t_W} > T_w, p_t \in P_p\} \tag{31}$$

Let $P = P_w$ represent the final filtered set of peaks which will be passed on for further analysis.

## 5.4 Peak Analysis

Several metrics are computed on the set of peaks $P$ which are used during peak selection. These metrics include:

- Basic Peak Metrics
  - Peak Boundaries: Left and right base indices, $p_{t_{BL}}$ and $p_{t_{BR}}$ respectively, found during the prominence computation in the initial peak finding (see Eq. 27).
  - Peak Width ($p_{t_W}$): Width at height $h_{eval}$, found during the width computation in the initial peak finding (see Eq. 31).
  - Peak Area ($p_{t_A}$): Computed using trapezoidal integration:

$$p_{t_A} = \frac{1}{2} \sum_{t=p_{t_{BL}}}^{p_{t_{BR}}-1} [x(t+1) - x(t)] \left[y_{corr}(t+1) + y_{corr}(t)\right] \tag{32}$$

    where $x(t)$ represents the time points corresponding to signal values. It should be noted that, for the GPPM and HGPPM algorithms, $y_{corr}$ is replaced with the Gaussian fitted curve of the peak $p_{t_G}(t)$ (see Eq. 37).

- Shape Analysis Metrics
  - Peak Symmetry ($p_{t_S}$): Normalized difference between left and right sides:

$$p_{t_S} = 1 - \frac{1}{y_{corr}(p_t)} \cdot \frac{1}{m} \sum_{i=1}^{m} \left| y_{corr}(p_{t_{BL}} + i) - y_{corr}(p_{t_{BR}} - i) \right| \tag{33}$$

    where $m = \min\{p_t - p_{t_{BL}}, p_{t_{BR}} - p_t\}$
  - Peak Skewness ($p_{t_\gamma}$): Third standardized moment of peak region:

$$p_{t_\gamma} = \frac{1}{n_p \sigma_p^3} \sum_{t=p_{t_{BL}}}^{p_{t_{BR}}} \left[y_{corr}(t) - \mu_p\right]^3 \tag{34}$$

    where $p_{t_N} = p_{t_{BR}} - p_{t_{BL}} + 1$ is the number of points, $p_{t_\mu}$ is the mean, and $p_{t_\sigma}$ is the standard deviation of the peak region computed from $y_{corr}(t)$
  - Gaussian Fit ($p_{t_G}(t|A, \mu, \sigma)$): Fitted using nonlinear least squares via SciPy's `curve_fit`:

$$p_{t_G}(t|A, \mu, \sigma) = A \cdot \exp\left(-\frac{(t-\mu)^2}{2\sigma^2}\right) \tag{35}$$

    The fit is found by computing:

$$\arg\min_{A,\mu,\sigma} \sum_{t=p_{t_L}}^{p_{t_R}} \left(y_{corr}(t) - p_{t_G}\left[t|A, \mu, \sigma\right]\right)^2 \tag{36}$$

– Normalized Residuals ($p_{t_G R}$): The normalized root mean square error between $p_{t_G}(t)$ and $y_{corr}(t)$:

$$p_{t_{GR}} = \frac{1}{y_{corr}(p_t)} \sqrt{\frac{1}{p_{t_N}} \sum_{t=p_{t_{BL}}}^{p_{t_{BR}}} \left[y_{corr}(t) - p_{t_G}(t|A,\mu,\sigma)\right]^2} \tag{37}$$

- Separation Metrics
  - Peak Resolution ($p_{t_R}$): For nearest neighboring peak $p_o \in P : p_o \neq p_t$:

$$p_{t_R} = \frac{2\,|x(p_t) - x(p_o)|}{p_{t_W} + p_{n_W}} \tag{38}$$

  - Peak Prominence ($p_{t_p}$): Height above highest minimum:

$$p_{t_p} = y_{corr}(p_t) - \max\{y_{corr}(p_{t_{BL}}), y_{corr}(p_{t_{BR}})\} \tag{39}$$

- Quality Score ($p_{t_Q}$): Combined metric for peak evaluation:

(40)

## 5.5 Peak Selection

For GPPM, only a single chromatogram may be passed in at a time. However, for CC and HGPPM, either a single chromatogram or multiple chromatograms can be passed in.

### 5.5.1 Independent Peak Selection

For the GPPM implementation, rather than directly using the latest peak time $p_t \in P$, the mean $\mu$ of $p_{t_G}(t|A,\mu,\sigma)$ which is optimized and returned by `curve_fit` is used to better approximate the true retention time:

$$p' = \arg\max_{p_t \in P} \{\mu : p_{t_G}(t|A,\mu,\sigma)\} \tag{41}$$

While CC is built to take in a hierarchy of chromatograms, if a single chromatogram is passed in, it will still return a result. In this case, the peak that is picked is the peak with the highest quality score:

$$p' = \arg\max_{p_t \in P} \{p_{t_Q}\} \tag{42}$$

Similarly, HGPPM is configured so that it will pick the peak (using the Gaussian mean as the time) of the peak with the highest $p_{t_Q}$ value:

$$p' = \left\{\mu : p_{t_G}(t|A,\mu,\sigma), p_t = \arg\max_{p_t \in P}\{p_{t_Q}\}\right\} \tag{43}$$

### 5.5.2 Hierarchical Peak Selection

For CC and HGPPM, when multiple chromatograms are provided in a hierarchy, the peak selection process traverses the hierarchy from bottom (the compounds with all building blocks but one being null) to top (compound with no null building blocks). Let $G = (V,E)$ be a directed acyclic graph representing the hierarchy where:

- $V$ is the set of vertices representing sequences
- $E$ is the set of directed edges $(u,v)$ where $u$ is an ancestor of $v$

- Each vertex $v \in V$ has:
  - A level $l(v)$ indicating the number of non-null elements
  - A set of initial peaks $P_v$ found during initial peak finding
  - A corrected signal $y_{corr,v}(t)$
  - A potentially selected peak time $p'_v$ (if one is chosen)

The graph satisfies:

$$\forall (u, v) \in E : l(u) > l(v) \tag{44}$$

Let $\tau$ be the time threshold parameter controlling valid peak selection for which all peaks must be greater than all of its descendants' picked peak times plus $\tau$. The hierarchical peak picking process is as such:

1. Terminal Vertex Peak Picking: For each terminal vertex $v$ where $l(v) = \min_{u \in V} l(u)$:

$$p'_v = \begin{cases} \arg\max_{p_t \in P_v} \{p_{t_Q}\} & \text{if } P_v \neq \emptyset \\ \text{undefined} & \text{otherwise} \end{cases} \tag{45}$$

2. Hierarchical Traversal: For each non-terminal vertex $u$ in ascending order of $l(u)$, define the set of descendants of $u$:

$$D_u = \{v : (u, v) \in E\} \tag{46}$$

Define the valid peaks $P_u^{valid}$ as:

$$P_u^{valid} = \{p_t \in P_u : p_t > p'_v + \tau, \forall\ v \in D_u\} \tag{47}$$

Then select the peak with maximum quality score from the valid peaks:

$$p'_u = \begin{cases} \arg\max_{p_t \in P_u^{valid}} \{p_{t_Q}\} & \text{if } P_u^{valid} \neq \emptyset \\ \text{undefined} & \text{otherwise} \end{cases} \tag{48}$$

For HGPPM specifically, we modify the peak selection to use the Gaussian mean:

$$p'_u = \begin{cases} \{\mu : p_{t_G}(t|A, \mu, \sigma), p_t = \arg\max_{p_t \in P_u^{valid}} \{p_{t_Q}\}\} & \text{if } P_u^{valid} \neq \emptyset \\ \text{undefined} & \text{otherwise} \end{cases} \tag{49}$$

The process terminates when the root vertex $r$ (where $l(r) = \max_{v \in V} l(v)$) has been processed, resulting in either a selected peak time $p'_r$ or undefined if no valid peak was found.

# 6 Split Tree Analysis

The Split Tree Analysis (STA) provides a hierarchical visualization and analysis framework for examining the success rates of peptide synthesis pathways based on chromatographic peak data.

## 6.1 Graph Construction

Let $G = (V, E)$ be a directed acyclic graph representing a Split-Tree where:

- $V$ is the set of vertices representing a peptide with one or more chromatograms associated with it.
- $E$ is the set of directed edges $(u, v)$ where $u$ is a parent of $v$
- Each vertex $v \in V$ has:
  - A level $l(v)$ indicating the position in the peptide

- A building block type $b(v)$
- A peak time $p'_v$ (if one exists)
- A success state $s_v \in \{\text{success}, \text{failure}, \text{undetermined}\}$

The graph satisfies the following properties:

$$\forall (u, v) \in E : l(v) = l(u) + 1 \tag{50}$$

$$\forall v \in V : l(v) \leq d_{max} \tag{51}$$

where $d_{max}$ is the number of positions in the peptide.

## 6.2 Analysis

The analysis works bottom-up through the tree. Let $D_v$ represent the set of descendants of vertex $v$, as defined in Eq. 46:

For terminal vertices (where $D_v = \emptyset$), the success state is determined by:

$$s_v = \begin{cases} \text{success} & \text{if } p'_v > p'_u + \tau \ \forall \ u \in D_v \\ \text{undetermined} & \text{if } p'_v \text{ is undefined} \\ \text{failure} & \text{otherwise} \end{cases} \tag{52}$$

though it should be noted that a separate success rate is computed as

$$r_v = \frac{|\{u \in D_v : (v, u) \in E \wedge s_u = \text{success}\}|}{|\{u \in D_v : (v, u) \in E\}|} \tag{53}$$

where $\tau$ is the time threshold parameter. For non-terminal vertices, the success state is determined by:

$$s_v = \begin{cases} \text{success} & \text{if } \exists \ u \in D_v : (v, u) \in E \wedge s_u = \text{success} \\ \text{failure} & \text{if } \forall \ u \in D_v : (v, u) \in E \wedge s_u = \text{failure} \\ \text{undetermined} & \text{otherwise} \end{cases} \tag{54}$$

The utility of STA can be extended given any chromatogram metric $m$ defined in Section 2.1, any chromatogram peak metric defined in Section 5.4 (except for the Gaussian Fit since it is not a scalar), or any user-defined values, by computing summary statistics across all chromatograms for vertex $v$:

$$\begin{aligned} \mu_m(v) &= \frac{1}{|M_v|} \sum_{i \in M_v} m(i) \\ \sigma_m(v) &= \sqrt{\frac{1}{|M_v| - 1} \sum_{i \in M_v} [m(i) - \mu_m(v)]^2} \end{aligned} \tag{55}$$

where $M_v$ is the number of chromatograms associated with vertex $v$ and $m(i)$ is the value of the the metric of the $i^{th}$ chromatogram. The nodes of the split-tree can be colored by $\mu_m(v), \sigma_m(v)$, or $\sigma^2_m(v)$ to allow for library-wide visualization.

The color intensity $c_v$ for any given non-terminal vertex border is computed as:

$$c_v = \frac{|\{u \in D_v : (v, u) \in E \wedge s_u = \text{success}\}|}{|\{u \in D_v : (v, u) \in E\}|} \tag{56}$$

where $c_v = 0$ maps to red (complete failure) and $c_v = 1$ maps to blue (complete success).

Each vertex is marked with an index relating it to a specific building block and it's body is colored by the chosen metric using a specified color map.

### 6.3 Layout

We've developed a graph layout specifically for efficient use of space, localization of vertex clusters, and visual clarity which employs a golden spiral fractal layout algorithm. For a set of $n$ vertices at depth $d$, their positions are determined by:

$$\begin{aligned} r_i &= \sqrt{\frac{i}{n}}(r_{max} - r_{min})f(d) + r_{min} \\ \theta_i &= i\phi \end{aligned} \tag{57}$$

where:

- $r_i$ is the radius for vertex $i$
- $\theta_i$ is the angle for vertex $i$
- $\phi = \pi(3 - \sqrt{5})$ is the golden angle
- $f(d) = 0.5^d$ is the depth scaling factor
- $r_{min}, r_{max}$ are the minimum and maximum radii

The final position $(x_i, y_i)$ of vertex $i$ relative to its parent is:

$$\begin{aligned} x_i &= r_i \cos(\theta_i) \\ y_i &= r_i \sin(\theta_i) \end{aligned} \tag{58}$$

## References